

DARLINGTON COUNTY AP, SC, USA

WMO: 723116

Lat: 34.450N

Lon: 79.883W

Elev: 59

StdP: 100.62

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03742

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-3.0	-13.9	1.1	-1.7	-12.1	1.3	0.5	11.0	12.9	9.7	12.7	1.8	350	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.6	24.6	33.9	24.2	32.8	24.0	26.8	31.9	26.1	31.3	25.5	30.7	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	29.3	24.6	19.7	28.4	24.0	19.0	27.8	84.1	32.0	80.9	31.3	77.9	30.8	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.9	7.7	6.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-9.8	37.3	2.2	1.7	-11.3	38.5	-12.6	39.5	-13.8	40.5	-15.4	41.7
			-10.3	27.5	2.1	1.1	-11.8	28.3	-13.0	29.0	-14.2	29.6	-15.7	30.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.8	8.5	12.4	17.5	22.3	26.3	26.9	26.5	23.7	17.7	11.8	8.4	(6)
(7)		DBStd	8.40	5.77	4.95	5.00	4.35	3.44	2.28	2.00	2.22	2.95	4.38	4.60	5.10	(7)
(8)		HDD10.0	380	136	81	33	2	0	0	0	0	0	2	32	94	(8)
(9)		HDD18.3	1483	360	277	191	67	9	0	0	0	3	65	201	311	(9)
(10)		CDD10.0	3096	35	39	108	226	381	489	525	511	410	241	86	44	(10)
(11)		CDD18.3	1157	1	2	8	41	131	239	267	252	163	46	4	2	(11)
(12)		CDH23.3	10566	3	12	84	421	1189	2314	2585	2302	1262	352	36	6	(12)
(13)	CDH26.7	4452	0	1	11	118	453	1067	1181	1049	483	87	3	0	(13)	
(14)	Wind	WSAvg	2.9	3.1	3.3	3.5	3.7	3.3	2.9	2.6	2.3	2.6	2.5	2.7	(14)	
(15)	Precipitation	PrecAvg	1164	83	80	90	74	80	120	131	145	116	88	74	81	(15)
(16)		PrecMax	1482	181	187	172	148	158	231	212	331	235	281	175	211	(16)
(17)		PrecMin	775	19	30	8	31	38	26	35	54	22	0	11	20	(17)
(18)		PrecStd	154	48	38	41	33	28	53	50	73	66	62	44	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	25.3	27.8	32.2	34.9	37.2	37.1	37.1	34.5	31.3	27.2	23.9	(19)
(20)			MCWB	17.4	18.4	18.2	19.7	22.6	24.5	25.5	25.2	23.3	23.4	19.7	19.2	(20)
(21)		2%	DB	21.1	22.2	26.1	29.8	32.7	35.0	34.8	34.9	32.7	28.9	24.0	22.1	(21)
(22)			MCWB	16.4	16.2	17.4	19.4	21.8	24.0	25.2	25.0	23.4	21.9	17.4	18.2	(22)
(23)		5%	DB	18.2	19.6	23.6	27.7	31.2	33.6	33.4	33.0	31.4	27.3	22.3	19.3	(23)
(24)			MCWB	15.0	14.8	15.9	18.2	21.4	23.7	25.1	24.6	23.2	20.0	16.9	16.6	(24)
(25)		10%	DB	16.2	17.5	21.3	26.1	29.0	32.3	32.4	32.2	29.5	25.1	20.1	17.4	(25)
(26)			MCWB	12.9	12.7	14.8	17.6	20.6	23.4	24.8	24.6	23.0	20.1	15.9	14.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	19.6	20.0	22.4	24.6	26.4	27.8	27.9	25.8	25.0	21.4	20.4	(27)
(28)			MCDWB	21.5	23.5	25.3	27.1	30.1	32.9	32.6	33.0	30.1	28.6	24.1	22.9	(28)
(29)		2%	WB	17.4	17.9	18.7	21.0	23.5	25.6	26.9	26.8	25.1	23.7	19.6	18.8	(29)
(30)			MCDWB	20.2	20.7	23.9	26.6	28.8	32.1	32.1	31.6	29.9	26.9	22.4	21.1	(30)
(31)		5%	WB	15.6	15.8	17.5	19.8	22.7	25.0	26.2	26.0	24.4	22.4	18.3	17.2	(31)
(32)			MCDWB	17.8	18.6	21.3	25.6	28.1	31.1	31.6	30.8	28.9	25.3	20.7	18.8	(32)
(33)		10%	WB	13.5	13.8	16.3	18.7	21.8	24.4	25.6	25.4	23.7	20.9	16.7	15.1	(33)
(34)			MCDWB	15.7	16.8	20.2	24.0	27.2	29.7	30.7	30.4	27.8	23.9	19.8	17.5	(34)
(35)	Mean Daily Temperature Range	MDBR	11.2	11.8	12.8	13.1	12.0	10.7	10.4	10.5	10.8	11.9	12.8	11.2	(35)	
(36)		5% DB	MCDBR	12.9	14.1	14.9	15.0	13.8	12.5	11.9	12.3	12.9	13.2	14.4	12.5	(36)
(37)			MCWBR	9.0	9.2	7.8	6.4	4.9	4.1	4.3	4.6	5.2	6.6	8.8	9.2	(37)
(38)		5% WB	MCDWB	11.1	12.7	12.4	12.4	11.2	10.8	10.5	10.7	10.3	9.8	11.3	11.0	(38)
(39)			MCWBR	8.6	9.6	7.7	6.2	4.6	4.2	4.5	4.7	4.5	5.6	8.4	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.326	0.361	0.406	0.467	0.493	0.544	0.515	0.432	0.391	0.339	0.313	(40)	
(41)		taud	2.500	2.469	2.402	2.265	2.147	2.117	1.998	2.069	2.290	2.373	2.475	2.547	(41)	
(42)		Ebn at Noon	893	922	912	881	828	803	760	775	828	836	858	877	(42)	
(43)		Edh at Noon	84	96	112	135	153	158	177	162	124	105	86	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.73	3.37	4.46	5.59	6.11	6.40	6.08	5.51	4.73	3.90	2.99	2.38	(44)	
(45)		RadStd	0.15	0.36	0.35	0.39	0.49	0.43	0.37	0.26	0.40	0.47	0.24	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (71 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+123	(47)

Nomenclature: See separate page